Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering & Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim: Perfrom Gray Level operation Image.**	
Experiment No: 7	**Date:12/08/2026**	**Enrollment No:92400133057**

Aim: **Perfrom Gray Level operation Image**

Github Link :- https://github.com/manavkasundra/DSIP

Code :

```
import cv2

import numpy as np

import matplotlib.pyplot as plt

import sys # Better practice for exiting scripts



# Read image directly in grayscale

src_image = cv2.imread('/Users/manavkasundra/DSIP/EXP7/img.jpg', cv2.IMREAD_GRAYSCALE)

# Check if image is loaded (Fixed indentation)

if src_image is None:

   print("Image not found! Please check the file path.")

   sys.exit()



# -------------------------

# 1. Image Negation

# -------------------------

negative_image = 255 - src_image



# -------------------------

# 2. Thresholding
```

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```
# -------------------------

# Pixels above 128 become 255 (white), others become 0 (black)

_, thresholded_image = cv2.threshold(src_image, 128, 255, cv2.THRESH_BINARY)



# -------------------------

# 3. Gamma Correction

# -------------------------

gamma = 2.0



# OPTIMIZATION: Using a Lookup Table (LUT) is much faster in OpenCV

# than doing floating-point math on every single pixel via NumPy.

look_up_table = np.array([((i / 255.0) ** (1.0 / gamma)) * 255

                          for i in np.arange(0, 256)]).astype("uint8")



gamma_corrected_image = cv2.LUT(src_image, look_up_table)



# -------------------------

# Display Images

# -------------------------

plt.figure(figsize=(15, 5))



plt.subplot(1, 4, 1)

plt.imshow(src_image, cmap='gray')
```

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```
plt.title("Original")

plt.axis('off')



plt.subplot(1, 4, 2)

plt.imshow(negative_image, cmap='gray')

plt.title("Negative")

plt.axis('off')



plt.subplot(1, 4, 3)

plt.imshow(thresholded_image, cmap='gray')

plt.title("Threshold")

plt.axis('off')



plt.subplot(1, 4, 4)

plt.imshow(gamma_corrected_image, cmap='gray')

plt.title(f"Gamma (y={gamma})")

plt.axis('off')



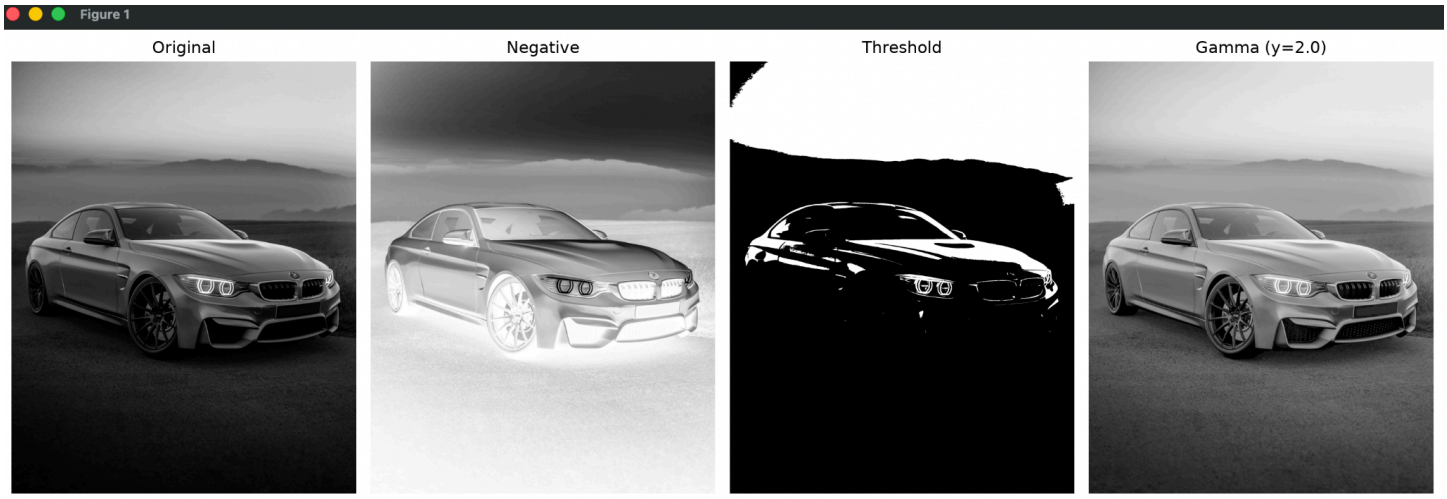
plt.tight_layout() # Prevents titles from overlapping

plt.show()
```

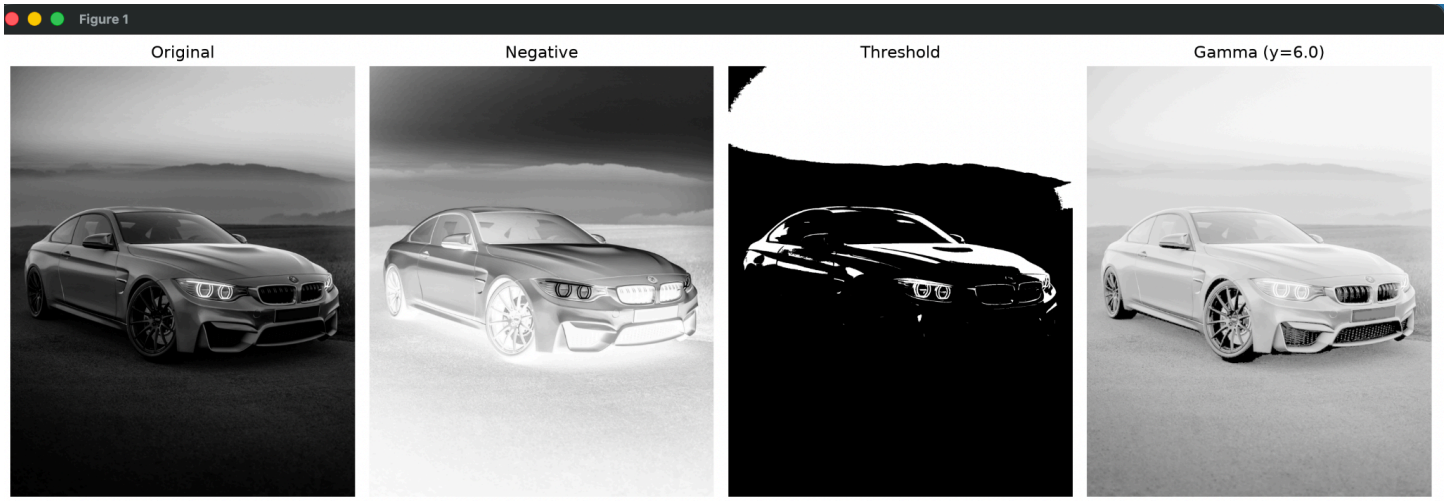
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Output:

Gamma = 2

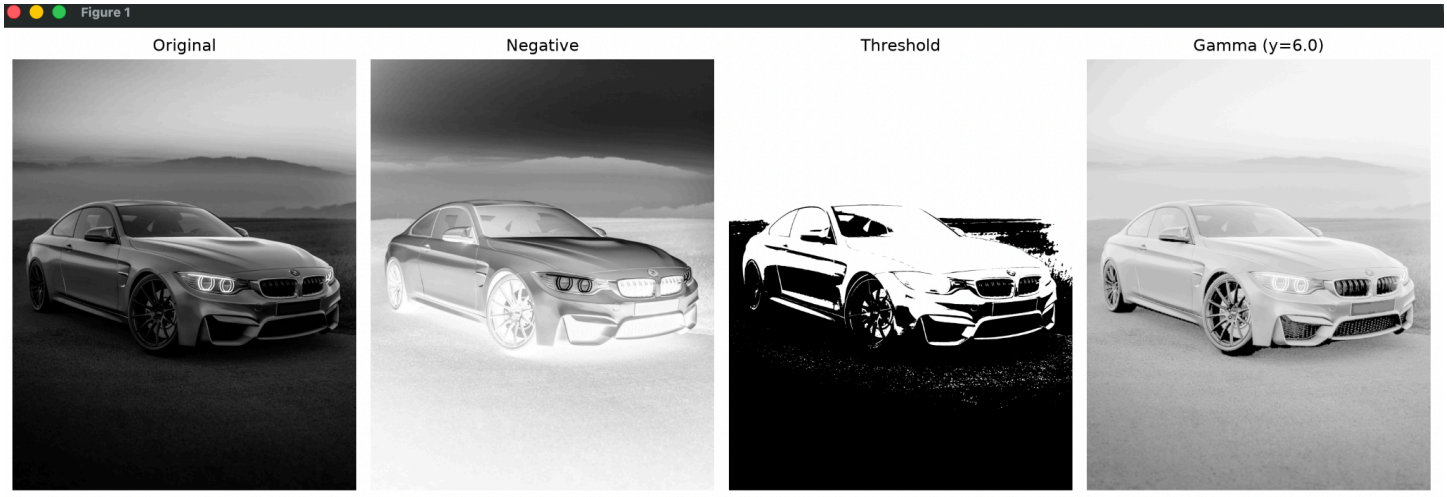


Gamma = 6

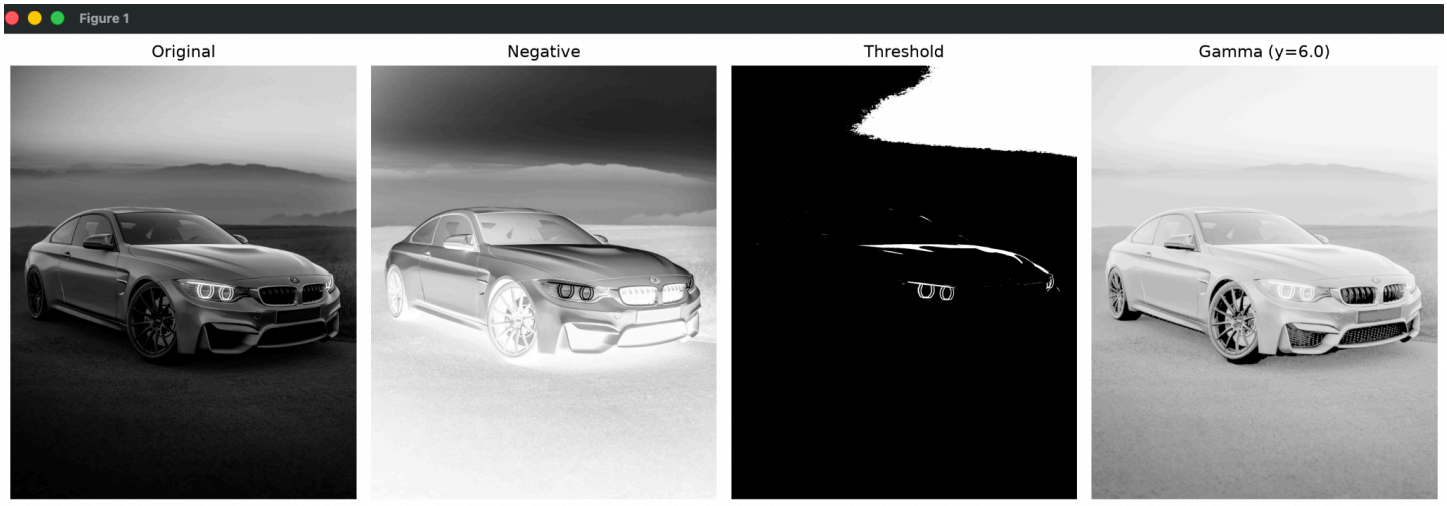


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Threshold = 30



Threshold = 200



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Conclusion :-

In this experiment, we successfully implemented basic gray-level image processing techniques, including negative transformation, thresholding, and gamma correction, using the OpenCV library.

- **Image Negation** effectively reversed the intensity levels, highlighting details in darker regions.
- **Thresholding** demonstrated how to segment an image into binary (black and white) regions based on pixel intensity, showing sensitivity to the chosen threshold value.
- **Gamma Correction** provided a non-linear adjustment, allowing us to control the luminance of the image, where higher gamma values noticeably brightened the output.

The results confirm that these fundamental operations are essential for image enhancement and pre-processing tasks in digital image processing.